Informatica

Informatica® Cloud Application Integration
November 2025

AI Agent for Customer Support Case Processing

Informatica Cloud Application Integration AI Agent for Customer Support Case Processing
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Preface

Use *AI Agent for Customer Support Case Processing* to learn how to automate the analysis of incoming customer emails, identify user intent, and trigger appropriate follow-up actions. This guide assumes that you have an understanding of the Email Connector, File Connector, Gemini Connector, Jira Connector, Pinecone Connector, and Zendesk Connector concepts.

CHAPTER 1

Introduction to AI Agent for Customer Support Case Processing recipe

The AI Agent for Customer Support Case Processing is an event-driven solution designed to automate the analysis of incoming customer emails by identifying user intent and triggering appropriate follow-up actions.

This versatile recipe is adaptable across various software environments and integrates seamlessly with systems, such as Zendesk, Jira, and Pinecone. It leverages an internal knowledge base powered by a configurable Large Language Model (LLM), with Gemini models supported by default. Users can customize the recipe to work with other GenAI platforms by establishing new connectors and replacing Gemini-specific calls as needed.

Designed for easy integration, the recipe enables seamless incorporation of customer-specific software. It can be easily extended by adding new agents and services, making it flexible to evolving business needs.

The agent processes emails based on three primary user intents, using distinct modes tailored to address each effectively:

- **Query Mode:** Handles customer inquiries seeking answers or problem resolution. The agent uses the LLM to generate automated responses and creates or updates a Zendesk ticket to track the issue.
- **Feedback Mode:** Manages user feedback on existing cases, such as acknowledgments, requests for additional support, or escalations. In this mode, the system closes the corresponding Zendesk ticket, marking the agent's task as complete.
- **Feature Request Mode:** Addresses requests for new features. The agent searches the linked Jira system for similar requests stored in the Pinecone database. If a match is found, it references the existing Jira ticket. Otherwise, it creates a new ticket for human follow-up. Newly created Jira tickets are indexed in Pinecone for future reference.

The workflow begins when the EmailConnection app triggers the recipe through an Email Listener Event. The agent extracts the email's subject and body, passing this content to the PlanningAgent process for intent analysis. Upon determining the user's intent, Query, Feedback, or Feature Request, the PlanningAgent generates a structured JSON object outlining the necessary steps. It then sequentially invokes specialized subagents to execute these steps.

Throughout the process, the system automates the following key actions:

- Creating or updating Zendesk tickets with the user's input and LLM-generated responses, with ticket status updated according to workflow progression.
- Creating new Jira tickets for feature requests or retrieving existing Jira tickets when similar issues are found in the Pinecone index.

- Sending timely email updates to users after each interaction, keeping them informed throughout the support process.

This intelligent, event-driven automation enhances support efficiency and improves customer satisfaction by combining AI-powered analysis with integration across popular support platforms. Although originally implemented using Gemini, the solution's flexible design allows easy adaptation to other GenAI technologies through appropriate connectors.

Prerequisites

To use the AI Agent for Customer Support Case Processing recipe, the following prerequisites must be met:

- Set up the Jira Project
- Create an index in Pinecone

Setting up the Jira Project

Before starting the project tasks, it is essential to create an appropriate Jira project for issue tracking and management. The project must include the following work types to ensure proper categorization and workflow of tasks:

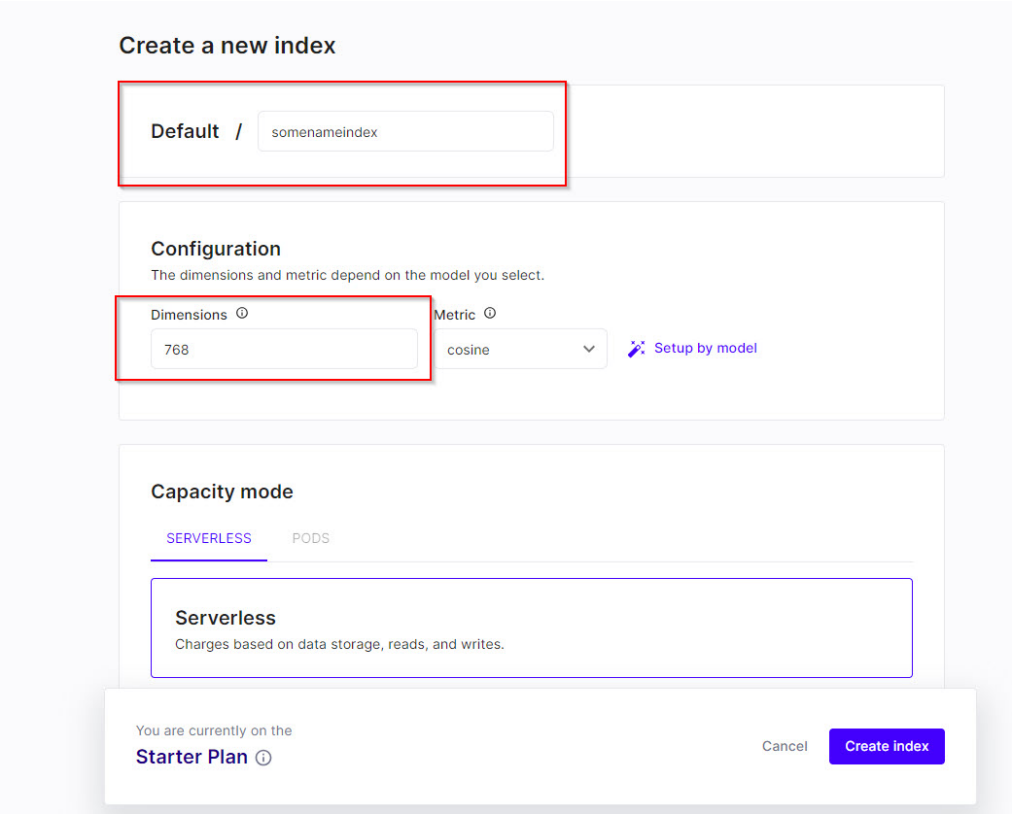
- New Feature
- Support
- Developer Escalation
- Bug

To simplify this process, you can use the existing Jira project template named **Customer Service Management**, which already contains all the required work types listed above. Using this template ensures consistency and saves time by avoiding configuring the project from scratch.

Creating an index in Pinecone

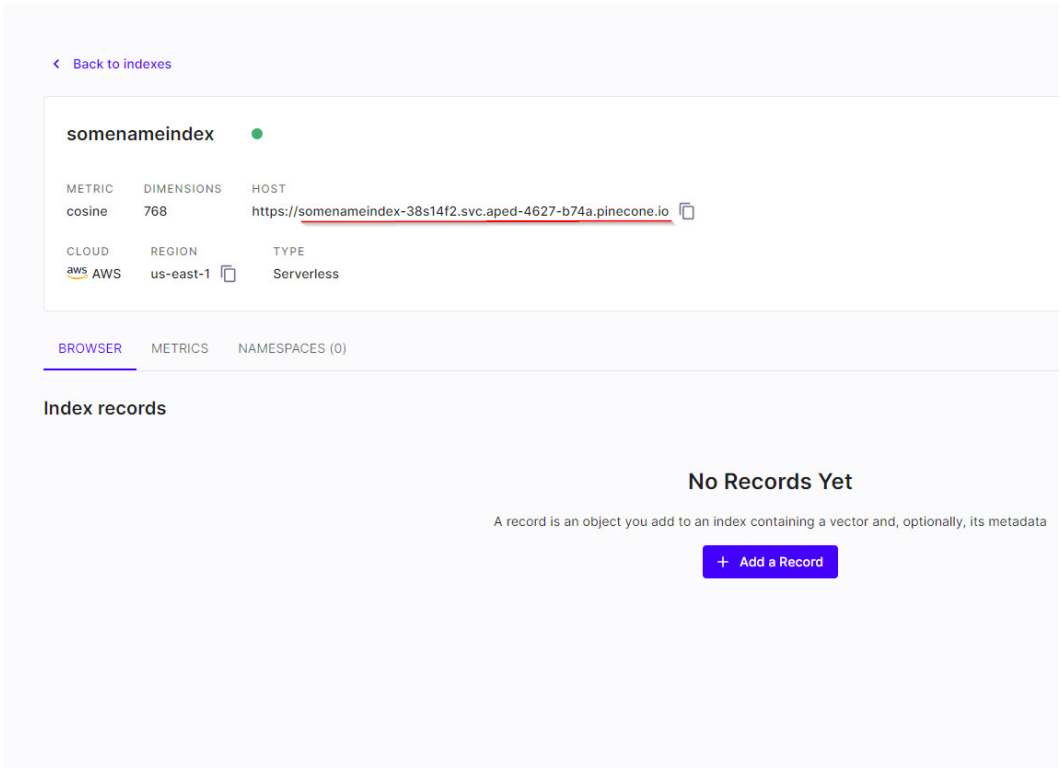
To ensure that the process works correctly, you must create an index in Pinecone and add text for the context. The context is then converted into vectors using the Get Embeddings process.

1. Open the **Create a new index** page in Pinecone.
2. In the **Default** field, enter an index name and in the **Dimensions** field, enter the value **768** as shown in the following image:



3. Click **Create index**.

After creating the index, you can use the **HOST** value without `https://` in the **Index_Host** input parameter as shown in the following image:

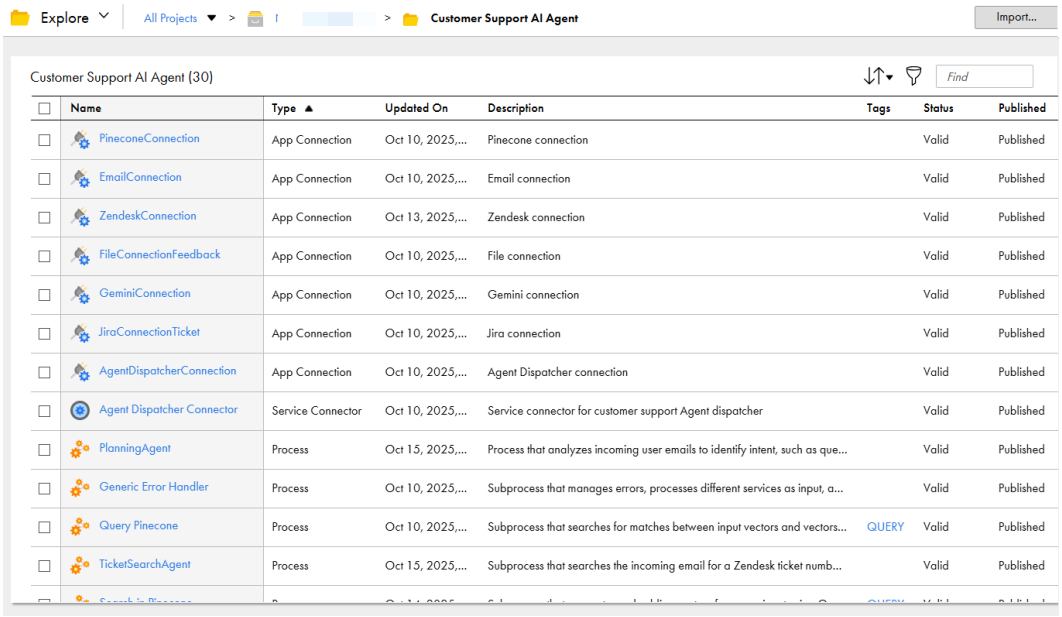

Back to indexes
somenameindex
METRIC
cosine
DIMENSIONS
768
HOST
https://somenameindex-38s14f2.svc.aped-4627-b74a.pinecone.io
CLOUD
AWS
REGION
us-east-1
TYPE
Serverless
BROWSER
METRICS
NAMESPACES (0)
Index records
No Records Yet
A record is an object you add to an index containing a vector and, optionally, its metadata
Add a Record

CHAPTER 2

AI Agent for Customer Support Case Processing recipe contents

The AI Agent for Customer Support Case Processing recipe contains a service connector, process objects, app connections, and processes.

The following image shows the assets that the AI Agent for Customer Support Case Processing recipe package contains:



Name	Type	Updated On	Description	Tags	Status	Published
PineconeConnection	App Connection	Oct 10, 2025,...	Pinecone connection		Valid	Published
EmailConnection	App Connection	Oct 10, 2025,...	Email connection		Valid	Published
ZendeskConnection	App Connection	Oct 13, 2025,...	Zendesk connection		Valid	Published
FileConnectionFeedback	App Connection	Oct 10, 2025,...	File connection		Valid	Published
GeminiConnection	App Connection	Oct 10, 2025,...	Gemini connection		Valid	Published
JiraConnectionTicket	App Connection	Oct 10, 2025,...	Jira connection		Valid	Published
AgentDispatcherConnection	App Connection	Oct 10, 2025,...	Agent Dispatcher connection		Valid	Published
Agent Dispatcher Connector	Service Connector	Oct 10, 2025,...	Service connector for customer support Agent dispatcher		Valid	Published
PlanningAgent	Process	Oct 15, 2025,...	Process that analyzes incoming user emails to identify intent, such as que...		Valid	Published
Generic Error Handler	Process	Oct 10, 2025,...	Subprocess that manages errors, processes different services as input, a...		Valid	Published
Query Pinecone	Process	Oct 10, 2025,...	Subprocess that searches for matches between input vectors and vectors...	QUERY	Valid	Published
TicketSearchAgent	Process	Oct 15, 2025,...	Subprocess that searches the incoming email for a Zendesk ticket numb...		Valid	Published

AI Agent for Customer Support Case Processing recipe assets

The following table lists the assets that the AI Agent for Customer Support Case Processing recipe package contains:

Asset Name	Asset Type	Description
UberPlanner_PO	Process object	Process object that contains custom attributes for the Uber Planner.
VectorMatchEntry_PO	Process object	Process object that provides all the information from the Pinecone vector matching.
AgentPlanner_PO	Process object	Process object that contains information about the agent's name and attribute references.
GenerationConfig_PO	Process object	Process object that creates the prompt request.
FileEmail_PO	Process object	Process object that contains email details.
JiraTicketDescription_PO	Process object	Process object that contains Jira ticket details.
Attributes_PO	Process object	Process object that contains email details, information required to create a Jira ticket, Zendesk ticket data, confidence level, and search parameters for querying data in Pinecone. This process object is used by another process object named UberPlanner_PO.
Agent Dispatcher Connector	Service connector	Service connector for customer support Agent dispatcher.
ZendeskConnection	App connection	Zendesk connection.
GeminiConnection	App connection	Gemini connection.
AgentDispatcherConnection	App connection	Agent Dispatcher connection.
JiraConnectionTicket	App connection	Jira connection.
FileConnectionFeedback	App Connection	File connection.
PineconeConnection	App connection	Pinecone connection.
EmailConnection	App connection	Email connection.
SummarizerAgent	Process	Subprocess that provides a concise, friendly summary of query agent's response to the user query, limited to under 300 tokens.

Asset Name	Asset Type	Description
TicketSearchAgent	Process	Subprocess that searches the incoming email for a Zendesk ticket number and updates the ticket if found. If the Zendesk ticket does not exist, it creates a new ticket. Additionally, it generates a Zendesk ticket link and logs the conversation history.
Search in Pinecone	Process	Subprocess that generates embedding vectors from user input using Gemini, searches for matching vectors in a specified Pinecone index based on input parameters, and returns the top matches with metadata, limited to the specified number of similar Pinecone vectors.
Create Jira Ticket	Process	Subprocess that creates a new Jira ticket.
Get Embeddings	Process	Subprocess that generates embedding vectors from the user input using Gemini.
Query Pinecone	Process	Subprocess that searches for matches between input vectors and vectors in base Pinecone specified by the input parameters Index_Host_Pinecone, and returns results with metadata.
PlanningAgent	Process	Process that analyzes incoming user emails to identify intent, such as query, feedback, or feature request, and routes the workflow accordingly.
Generic Error Handler	Process	Subprocess that manages errors, processes different services as input, and returns appropriate error responses based on the service type and the error encountered, ensuring effective error handling and recovery.
Upsert To Pinecone Index	Process	Subprocess that creates vectors using Gemini AI and saves them in an Pinecone index.
FeedbackAgent	Process	Subprocess that updates a Zendesk ticket by marking it as feedback-related and sends a follow-up email to thank the user for their feedback.
Gemini Agent Dispatcher	Process	Subprocess that calls the sub-agents based on the query prompt and dispatches the workflow to the appropriate agent at run time.
EvaluationAgent	Process	Subprocess that evaluates the adequacy of the Query Agent's response to the user's question based on five criteria—Correctness, Relevance, Clarity, Completeness, and Style and Tone—and returns a confidence level.
QueryAgent	Process	Subprocess that generates a response based on the outcome of the Search in Pinecone subprocess, using either articles stored in Pinecone or its own knowledge base.

Asset Name	Asset Type	Description
FeatureRequestAgent	Process	Subprocess that searches for similar JIRA tickets related to a similar feature mentioned in the email. If no matching tickets are found, it creates a new Jira ticket and sends a follow-up email to the user.
ConfidenceLevelHandlerAgent	Process	Subprocess that depending on the ConfidenceLevel value, either sends a response generated by the Query Agent or Summarizer Agent, or searches for similar Jira tickets related to a similar feature mentioned in the email. If no matching tickets are found, it creates a new Jira ticket and sends a follow-up email to the user.

CHAPTER 3

Using the AI Agent for Customer Support Case Processing recipe

To use the AI Agent for Customer Support Case Processing recipe, you must perform the following steps manually:

1. Copy and access the recipe.
2. Publish the Agent Dispatcher Connector service connector.
3. Configure and publish the AgentDispatcherConnection connection.
4. Configure and publish the Zendesk connection.
5. Configure and publish the Jira connection.
6. Configure and publish the Gemini connection.
7. Configure and publish the Pinecone connection.
8. Configure and publish the Email connection.
9. Configure and publish the File connection.
10. Configure and publish the processes.
11. Invoke the process.

Step 1. Copy and access the recipe

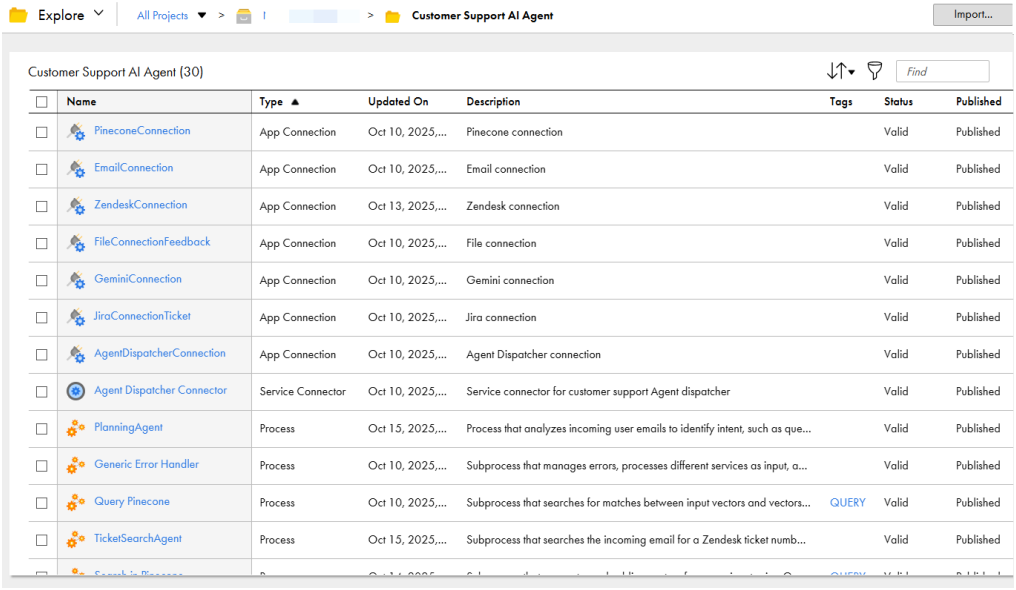
Copy the pre-configured assets in the recipe to a separate project or folder.

1. Open the **AI Agent for Customer Support Case Processing** recipe, and then click **Use**.
2. Select the location where you want to copy the recipe, and then click **Continue**.
3. In the **Copying the recipe** dialog box, click **OK**.

   It might take some time for the recipe to get copied. You will receive a notification when the recipe is ready for use.
4. After the recipe is copied, click **Explore** to access the recipe content.

5. Navigate to the project or folder where you copied the recipe or enter the recipe name in the **Find** box.

   All the assets in the recipe are displayed as shown in the following image:



Step 2. Publish the Agent Dispatcher Connector service connector

You can publish the **Agent Dispatcher Connector** service connector from the assets page.

To publish the **Agent Dispatcher Connector** service connector, click **Actions** in the row that contains the service connector and select **Publish**.

Step 3. Configure and publish the Agent Dispatcher connection

Configure the connection type in the AgentDispatcherConnection connection, and then publish the connection.

1. Open the **AgentDispatcherConnection** connection.
2. From the **Type** list, select **Agent Dispatcher Connector** from the recipe folder.
3. From the **Runtime Environment** list, select **Cloud Server or any Secure Agent**.
4. Save and publish the connection.

Step 4. Configure and publish the Zendesk connection

Configure the email address, API token, and instance domain to connect to Zendesk, and then publish the ZendeskConnection connection.

1. Open the **ZendeskConnection** connection.
2. From the **Type** list, select **Zendesk**.
3. From the **Runtime Environment** list, select **Cloud Server or any Secure Agent**.
4. In the **Connection Properties** section, enter values for the following properties:

Property	Description
Email	Zendesk user's email address to authenticate requests.
Api_Token	Zendesk API token to authenticate requests. You can generate this value on the Zendesk Admin page.
Instance_Domain	The instance domain of Zendesk. Enter the instance domain in the following format: `https://{domain}.zendesk.com`

5. Save and publish the connection.

Step 5. Configure and publish the Jira connection

Configure the instance domain, user name, and authentication token to connect to the Jira instance, and then publish the JiraConnectionTicket connection.

1. Open the **JiraConnectionTicket** connection.
2. From the **Type** list, select **Jira**.
3. From the **Runtime Environment** list, select **Cloud Server or any Secure Agent**.
4. In the **Connection Properties** section, enter values for the following properties:

Property	Description
InstanceDomain	The URL to access the Jira instance. For example, `https://<<name>>.atlassian.net`
UserName	The user name of the Jira account.
AuthToken	The Jira API token that you generated from **Manage Account** > **Security** > **Create and manage API tokens**.

5. Save and publish the connection.

Step 6. Configure and publish the Gemini connection

Configure the API key to authenticate the Gemini connection, and then publish the GeminiConnection connection.

1. Open the **GeminiConnection** connection.
2. From the **Type** list, select **Gemini**.
3. From the **Runtime Environment** list, select **Cloud Server or any Secure Agent**.
4. In the **Connection Properties** section, enter the API key in the **API_Key** property. The **API_Key** property authenticates Gemini connection requests.
5. Save and publish the connection.

Step 7. Configure and publish the Pinecone connection

Configure the API key to authenticate the Pinecone connection, and then publish the PineconeConnection connection.

1. Open the **PineconeConnection** connection.
2. From the **Type** list, select **Pinecone**.
3. From the **Runtime Environment** list, select **Cloud Server or any Secure Agent**.
4. In the **Connection Properties** section, enter the API key in the **API_Key** property. The **API_Key** property authenticates Pinecone connection requests.
5. Save and publish the connection.

Step 8. Configure and publish the Email connection

Configure the authentication details, host and port of the email server in the EmailConnection connection, and then publish the connection.

1. Open the **EmailConnection** connection.
2. From the **Type** list, select **IICS Cloud Application Integration Email Service (Licensed for use)**.
3. From the **Runtime Environment** list, select **Cloud Server or any Secure Agent**.

4. From the **Authentication Type** list, select **Password** or **OAuth** as needed. Based on the authentication type selected, perform one of the following steps:
   - For **Password** authentication, enter values for the following properties in the **Connection Properties** section:

Property	Description
Authentication	Select **Enable**. Email Connector authenticates the user name and password that you enter in the email connection properties.
User Name	User name to log in to the email server. The user name is either the account name or the email address that is used to send the email with the synchronization results. For example: `notifyme@mydomain.com`
Password	Password for the email address. Set an API key for your email account. For information about creating an API key, see Create API credentials.
Security	Select **SSL** for the Email connection to use the SSL protocol.
Connection Timeout	The connection timeout in milliseconds. The default is 30000 milliseconds.

Configure the following common properties on the connection creation page:

Property	Description
Host	Email server's DNS name, such as `mail.mydomain.com`, or an IP address, such as `192.168.1.1`.
Port	Port for communication between the Process Server and the email server. Default is **25**.
Test Email Address	The email address that receives test mails from this connection.
From Email Address	The email address that appears in the From field for emails sent from this connection.

- For **OAuth** authentication, enter values for the following properties in the **Connection Properties** section:

Property	Description
Authorization URL	Enter the OAuth authorization URL for the email service that is used to authorize the user request. For example: `https://login.microsoftonline.com/xxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxx/oauth2/v2.0/authorize`
Token Request URL	Enter the OAuth token request URL that handles token requests. For example: `https://login.microsoftonline.com/xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx/oauth2/v2.0/token` The refresh token expires in 90 days. The user must authenticate again and publish the connection before the token expires.
Client ID	Specify the identifier value from the OAuth provider.
Client Secret	Enter the client secret to connect to the email application.
Scope	Specify the scope. The scope in OAuth authentication limits an application's access to a user's account. You can select multiple scopes for a single client. To enter multiple scopes, separate each value with a space. For a Microsoft Outlook email account, enter the following scope: `https://outlook.office.com/SMTP.Send offline_access`
Authorization Status	Indicates the current status and the last time that authorization was completed.
Authorize Access	Click **Authorize** to initiate the authorization workflow using OAuth.

Configure the following common properties on the connection creation page:

Property	Description
Host	Email server's DNS name, such as `mail.mydomain.com`, or an IP address, such as `192.168.1.1`.
Port	Port for communication between the Process Server and the email server. Default is **25**.
Test Email Address	The email address that receives test mails from this connection.
From Email Address	The email address that appears in the From field for emails sent from this connection.

5. Save and publish the connection.

Step 9. Configure and publish the File connection

Configure the file directory and specify the action to be taken if a file already exists, and then publish the FileConnectionFeedback connection.

1. Open the **FileConnectionFeedback** connection.
2. From the **Runtime Environment** list, select the Secure Agent.
3. On the **Event Targets** tab, in the **Directory** field, enter the directory to store the files. In the **File Exists** field, select **Append**.
4. Save and publish the connection.

Step 10. Configure and publish the processes

The process is triggered when a user sends an email to a specified email address. The email's subject and body are passed to the **TicketSearchAgent** process. Upon completion, a prompt is generated to assess the intent of the email. The **Gemini Agent Dispatcher** process then determines and executes the subsequent steps based on the identified intent.

If the email is too short or unclear to determine intent, the process follows an alternate path where the clarity check fails. In such cases, the system sends an email back to the user explaining the ambiguity and requests a more detailed description of the issue.

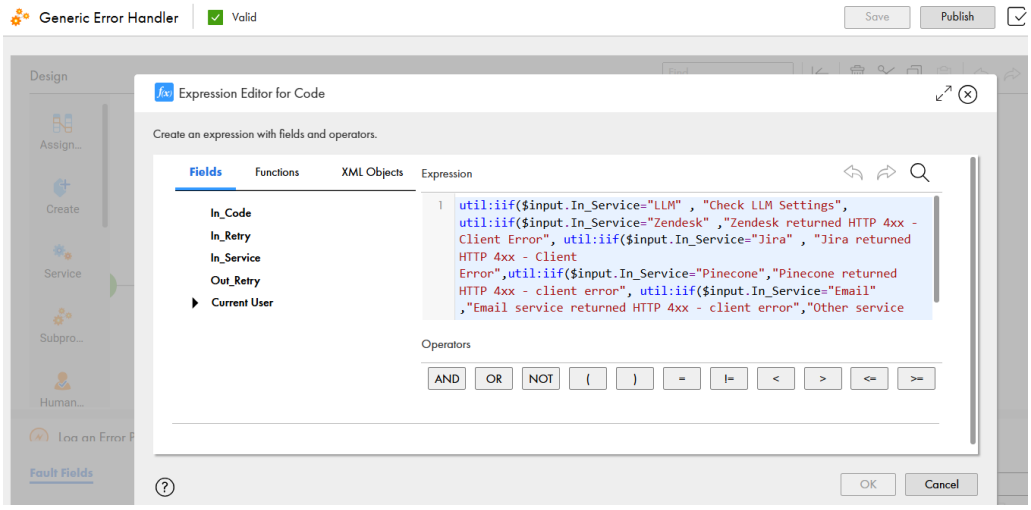
Additionally, the process includes error handling managed by the **Generic Error Handler** process to ensure robust handling of unexpected issues.

1. To publish the following processes, click **Actions** in the row that contains the process and select **Publish**.
   a. Generic Error Handler
   b. Get Embeddings
   c. Gemini Agent Dispatcher
   d. FeedbackAgent
   e. EvaluationAgent
   f. SummarizerAgent
   g. TicketSearchAgent
   h. Upsert to Pinecone Index
   i. Search in Pinecone
   j. Create Jira Ticket
   k. QueryAgent
   l. FeatureRequestAgent
   m. ConfidenceLevelHandlerAgent
2. Open the **Query Pinecone** process.
3. On the **Temp Fields** tab of the **Start** step, enter value for the **Tmp_Top_K** field and set **Tmp_Retry** to 0. The **Tmp_Top_K** field specifies the TopK value for Pinecone, which is the number of vectors retrieved.
4. Save and publish the process.
5. Open the **PlanningAgent** process.

6. On the **Temp Fields** tab of the **Start** step, enter values for the following fields:
   - **Tmp_Model_Version**. Model version to be used in all LLM calls.
   - **Tmp_Support_Company_Name**. Name of the company using the agent.
   - **Tmp_Assignee_Email**. Email address of the person assigned to the Zendesk ticket.
   - **Tmp_Output_Dimensionality**. Defines the precision of data stored in Pinecone based on the vector dimensions.
   - **Tmp_Embedding_Model**. Model used for embedding.
   - **Tmp_Jira_Pinecone_Index_Host**. Host address of the Pinecone index used to identify similar Jira tickets.
   - **Tmp_Jira_Project_Key**. Key of the Jira project where tickets are created.
   - **Tmp_RAG_Article_Cutoff**. Minimum Pinecone similarity score required for entries to be included in the RAG (Retrieval-Augmented Generation) process.
   - **Tmp_Jira_Cutoff**. Minimum Pinecone similarity score needed for entries to be considered as similar Jira tickets.
   - **Tmp_Similar_Pinecone_Vector_Number**. Number of similar Pinecone entries sent to the agent.
   - **Tmp_RAG_Pinecone_Index_Host**. Host address of the Pinecone index used for RAG.
   - **Tmp_Retry**. Set to 0 to reset the counter for failed external service calls.
   - **Tmp_NoClarityRecipients**. Email recipients who receive notifications when the user's email is unclear and the agent cannot respond.
7. Save and publish the process.

Additional information

- To enhance the recipe's functionality with calls to additional external services, users can utilize the **Generic Error Handler** process to catch faults and retry failed calls. To fully leverage the Generic Error Handler's capabilities, the user must update the formula that determines the **Code** field value in the **Log an Error** step. Specifically, the formula must include the value of the input field **In_Service** to clearly indicate which service returned the error code, similar to how all service names are referenced in the existing formula, along with the appropriate error message as shown in the following image:



- The PlanningAgent prompt must be Base64-encoded due to XML elements present in the prompt. Otherwise, parsing issues would occur. If the user wants to enhance the agent's capabilities by adding

new agents or additional paths for the planner, they must first decode the prompt into plaintext, make the desired modifications, and then re-encode it in Base64.

- To add more inputs for subagents other than the PlanningAgent, these inputs must be added to the **Attributes_PO** process object. This process object is part of the **AgentPlanner_PO** process object, which is passed as input to the Dispatcher action within the **Agent Dispatcher Connector** service connector and to all related subagents.

Step 11. Invoke the process

The agent is designed to be triggered using the email listener feature of the **EmailConnection** connection. To activate the agent, an email must be sent to the email address specified as the user name in the **EmailConnection** connection. When the connection successfully reads the email content, it initiates the **PlanningAgent** process, passing the email body, subject, and sender as inputs.

Examples

The following examples demonstrate how the system processes different types of user emails, such as inquiries, feedback, and feature requests, showcasing its automated processes and seamless integration with support and issue-tracking platforms.

Example 1: Handling User Queries

When a user sends a question or reports an issue via email, the AI agent responds automatically and creates a Zendesk ticket. If the user follows up, the agent provides additional replies, updating the ticket accordingly. After the issue is resolved, the user confirms resolution, and the agent classifies the final message as feedback, updating the ticket status accordingly.

Step-by-step process for user queries:

1. User sends an email with a query.
2. The AI agent sends an automated reply.
3. A Zendesk ticket is created with the query details.
4. The user replies through email.
5. The agent responds to the follow-up questions.
6. The Zendesk ticket is updated with latest information.
7. The user resolves the issue following the agent's instructions, for example by reviewing logs, and sends a confirmation reply.
8. The agent classifies this message as feedback, prefixes the ticket title and email subject with `[FEEDBACK]`, and sets the ticket status to Pending, awaiting any further action, such as reopening or closing the ticket.
9. The agent sends an acknowledgment of the feedback.
10. The Zendesk ticket status reflects the feedback classification.

Example 2: Processing Feature Requests

When a user requests a new feature by email, the agent acknowledges the request, creates a Zendesk ticket that is immediately marked as Closed, and also generates a Jira ticket for detailed tracking. For subsequent requests of the same feature, the agent links to the existing Jira ticket to prevent duplication, while still opening new Zendesk tickets for user reference and follow-up.

Step-by-step process for feature requests:

1. User sends a feature request email. For example, a custom workflow wizard in Informatica Customer 360.
2. The agent automatically acknowledges receipt.
3. A Zendesk ticket is generated and immediately marked as `Closed` since the feature request will be managed through a Jira ticket.
4. A Jira ticket is created to track and manage the feature request.
5. For similar requests later, the agent avoids creating duplicate Jira tickets by referencing the existing Jira ticket.
6. The agent includes a link to the existing Jira ticket in its response instead of opening a new one.
7. Each similar request also results in a new Zendesk ticket, which remains `Open` for the user to review the linked Jira ticket to verify if it aligns with their request or dismiss false positives.